SIMATS ENGINEERING

CO2-ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1706

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Furthose samreen B (192511164) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Furthose samreen b

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
    A
   /  \
  B    C
 / \  / \
D   E F   G
```

Task: Starting from node A, perform a Breadth-First Search (BFS). Complete the following table.

Iteration	Queue Before Visiting	Visited Node	Queue After Expansion
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

Questions

1. Write the BFS traversal order.

Answer:

A → B → C → D → E → F → G

2. Show the queue contents after each iteration.

Answer:

Iteration	Queue
1	B, C
2	C, D, E

3	D, E, F, G
Iteration	**Queue**
4	E, F, G
5	F, G
6	G
7	Empty

2. Depth-First Search (DFS)

Using the same graph,

```
      A
    /   \
   B     C
  / \   / \
 D   E F   G
```

Perform Depth-First Search (DFS) starting from A.

Answer

DFS Traversal Order:

A → B → D → E → C → F → G

Iteration	Stack Before Visiting	Visited Node	Stack After Expansion
1	A	A	C, B
2	C, B	B	C, E, D
3	C, E, D	D	C, E
4	C, E	E	C
5	C	C	G, F
6	G, F	F	G

7	G	G	Empty

3. 8-Puzzle using Breadth-First Search (BFS)

Initial State

1 3 6

5 0 2

4 7 8

Goal State

1 2 3

4 5 6

7 8 0

Note: The given initial state **cannot reach the given goal state** because it is an **unsolvable 8-puzzle configuration** (it has odd inversion parity).

Therefore, **BFS will not reach the goal state**, and the requested iteration table **cannot be completed for this input**.

Questions

1. Which node is expanded in each iteration?

Answer:

BFS expands the states in **level order**, removing the first state from the queue and expanding all its valid neighboring states until the goal is found.

For the given input, the goal state is **never reached** because the puzzle is unsolvable.

2. How many states are generated in total?

Answer:

Since the puzzle is **unsolvable**, BFS explores **all reachable states** from the initial configuration.

For the 8-puzzle, there are **181,440 reachable states** in one parity class.

3. Write the complete solution path from the initial state to the goal state.

Answer:

No solution exists because the given initial state cannot reach the goal state.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Answer:

- BFS explores states **level by level**.
- Each move has the **same cost**.
- The first time the goal is reached, it is guaranteed to be the **minimum number of moves**.
- Therefore, BFS always finds the **shortest path** in an unweighted graph.